

SECTION 32 11 23
AGGREGATE BASE COURSES (EMEA)

TABLE OF CONTENTS

PART 0 PROJECT-SPECIFIC AMENDMENTS

PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 QUALITY ASSURANCE
- 1.3 SUBMITTALS
- 1.4 PRODUCT HANDLING

PART 2 PRODUCTS

- 2.1 GENERAL
- 2.2 COARSE AGGREGATE
- 2.3 FINE AGGREGATE
- 2.4 COMPOSITE MIXTURE
- 2.5 PRIME ASPHALT
- 2.6 SUB-BASE MATERIAL

PART 3 EXECUTION

- 3.1 PREPARATION OF SUBGRADE
- 3.2 LAYING
- 3.3 ROLLING AND COMPACTING
- 3.4 ALLOWABLE TOLERANCES

PART 0 PROJECT-SPECIFIC AMENDMENTS

The following amendments to this Section apply only to this Project:

- 2.1 (A): Replace: ****LOCAL**** with "Austrian"
- 2.1 (A): Replace: (****LOCAL****) with "(Richtlinien und Vorschriften für den Straßenverkehr)"
- 2.2: Replace: ****LOCAL**** with "Austrian"
- 2.3: Replace: ****LOCAL**** with "Austrian"
- 2.4: Replace: ****LOCAL**** with "Austrian"
- 2.5: Replace: ****LOCAL**** with "Austrian"

PART 1 GENERAL

1.1 DESCRIPTION

- A. Work included: Provide crushed stone base (with prime) constructed on the compacted subgrade where shown on the Drawings, as specified herein, and as needed for a complete and proper installation.
- B. Related work:
 - 1. Drawings and the Construction Agreement, including any Amendments thereto, and Sections in Division 01 apply to this Section.
 - 2. Section 31 22 00 - Grading.
 - 3. Section 32 12 16 - Asphaltic Paving.

1.2 QUALITY ASSURANCE

- A. Use adequate numbers of skilled workmen who are thoroughly trained and experienced in the necessary trades and who are completely familiar with the specified requirements and the methods needed for proper performance of the work of this Section.

1.3 SUBMITTALS

- A. Comply with pertinent provisions of Section 01 33 00 - Submittal Procedures.
- B. Certificates, signed by materials producer, stating that materials meet the specified requirements.

1.4 PRODUCT HANDLING

- A. Comply with pertinent provisions of Section 01 60 00 - Product Requirements.

PART 2 PRODUCTS

2.1 GENERAL

- A. All materials and products used shall comply with pertinent sections of the ****LOCAL**** Austrian Department of Transportation's (****LOCAL****) (Richtlinien und Vorschriften für den Straßenverkehr) Design Standards & Specifications, latest edition with latest revisions and supplements.

2.2 COARSE AGGREGATE

- A. All materials, products, and composition of mixture used shall comply with pertinent sections of the Austrian ****LOCAL**** Design Standards & Specifications, latest edition with latest revisions and supplements.

2.3 FINE AGGREGATE

- A. All materials, products, and composition of mixture used shall comply with pertinent sections of the Austrian ****LOCAL**** Design Standards & Specifications, latest edition with latest revisions and supplements.

2.4 COMPOSITE MIXTURE

- A. All materials, products, and composition of mixture used shall comply with pertinent sections of the Austrian ****LOCAL**** Design Standards & Specifications, latest edition with latest revisions and supplements.

2.5 PRIME ASPHALT

- A. All materials, products, and composition of mixture used shall comply with pertinent sections of the Austrian ****LOCAL**** Design Standards & Specifications, latest edition with latest revisions and supplements.

2.6 SUB-BASE MATERIAL

- A. Sub-base material is defined as granular material to be used where indicated on the drawings or where specified herein, or as otherwise approved by the Owner's Geotechnical Engineer as select fill.
- B. This granular stone material grading shall be:

Sieve Size	% by Mass Passing
63 mm	100
31.5 mm	80 to 99
16 mm	55 to 85
8 mm	35 to 65
4 mm	22 to 50
2 mm	15 to 40
1 mm	10 to 35
0.5 mm	0 to 20
0.0063 mm	0 to 7

PART 3 EXECUTION

3.1 PREPARATION OF SUBGRADE

- A. Following topsoil removal and excavation to the formation level as necessary to allow a minimum 250 mm layer of select material to top of subgrade.
- B. Proposed formation level shall be the top surface of capping. Where no capping is required formation shall be the top surface of earthworks at underside of sub-base unless otherwise shown on the drawings.
- C. Proof Roll all areas to receive crushed stone paving.
 - 1. Make not less than six passes over the full area, using a 5,000 kg or higher machine.
- D. Sub-base Compaction, at a moisture content in the range optimum - 2 percent or + 1 percent and without drying out or segregation.
- E. The full thickness of the sub-base shall be continued for a distance of 300 mm beyond the limits of the kerb beam.
- F. Remove all soft, unstable or unsuitable material that will not compact readily.
 - 1. Soft spots shall be defined as areas where the soil does not meet the minimum shear strength requirement and should be identifiable when the soil undergoes excessive deformation during proof rolling with the roller. The extent of the soft area shall be determined by inspection during proof rolling. The Contractor shall advise the Engineer of any soft or otherwise unsuitable areas of sub-formation in excess of 1.0 m².
 - 2. Remove to full depth of unsuitable material, or to a depth of 1000 mm, whichever is less.
 - 3. Replace with satisfactory subbase material, spread and compacted in 150 mm layers.
 - 4. Soft spots shall be excavated and backfilled. Records of pavement foundation inspections and remediation measures carried out shall be kept by the Contractor and made available to the Engineer.
- G. Fill all holes, ruts or depressions which develop in the subgrade with approved select fill upper 250 mm of subgrade material, bringing subgrade to indicated line and grades.
- H. Compact subgrade using suitable construction procedures to provide not less than 95 percent Standard Proctor Maximum Dry Density.
- I. Seal roll the subgrade surface with a steel wheel roller, sealing the surface against excessive water infiltration.
- J. The Contractor shall undertake lightweight deflectometer testing (LWD) to ensure a minimum foundation surface modulus stiffness value is achieved prior to the placement of any subsequent asphalt layers.
- K. If the subgrade material is poor with a less than 20 kN/m² bearing pressure a geogrid or other form of ground stabilisation shall be used.

3.2 LAYING

- A. Place aggregates using spreader boxes or other approved spreaders uniformly on one operation.
- B. Take care to avoid segregation of the fine from the coarse aggregate during handling, spreading or shaping operations.
- C. Mix, while at proper moisture, with motor grader or other equipment and maintain the required section and grade until thoroughly compacted.

3.3 ROLLING AND COMPACTING

- A. Perform using 3-wheel steel wheel roller weighing not less than 10,000 kg, tandem roller weighing at least 8,000 kg, or other rollers approved by the Engineer.
- B. Start rolling at edges and proceed toward the center, continue rolling until aggregates are firmly keyed or set.
- C. When initial compaction is completed, should voids remain, place fine aggregates on the surface in an amount only sufficient to fill the voids.
- D. Broom, wet and roll until coarse aggregate is set, bonded and thoroughly compacted for full width and depth.
- E. All soft areas which develop during compaction shall be removed and replaced with approved materials and compacted in layers not exceeding 225 mm in depth.
- F. Where any completed area of sub-base is not immediately covered with the next construction layer, it shall be protected by a membrane of 500 Grade plastic sheeting with 300mm laps set to prohibit ingress of moisture. Alternatively a bituminous emulsion may be used as specified in EN 13808.
- G. No construction traffic shall be allowed on the completed sub-base, other than that required to lay the road base.

3.4 ALLOWABLE TOLERANCES

- A. Finished surfaces at each stage of road construction shall not vary from the levels described in the Contract by more than the following permissible tolerances:

Surface	Permissible Tolerance (mm)
Formation and sub-base	+10, -30

- B. Deviations: Correct by removing materials, replacing with new materials, and reworking or recompacting as required.

END OF SECTION

REVISION HISTORY

Version	Date	Description and location of change
0.1	August 2024	Initial version – BOD Draft submission
0.2	September 2024	BOD Final Submission
0.3	February 2025	Draft IFC Updates
1.0	12 th of March 2025	Final IFC Issue